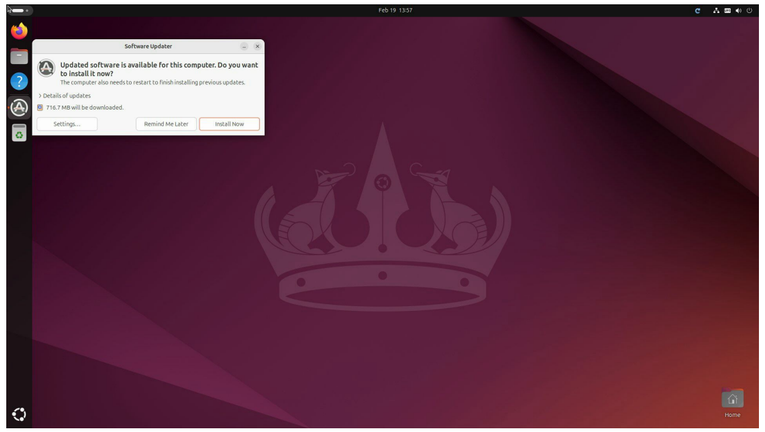
OCR-PrJ

19/02/2026

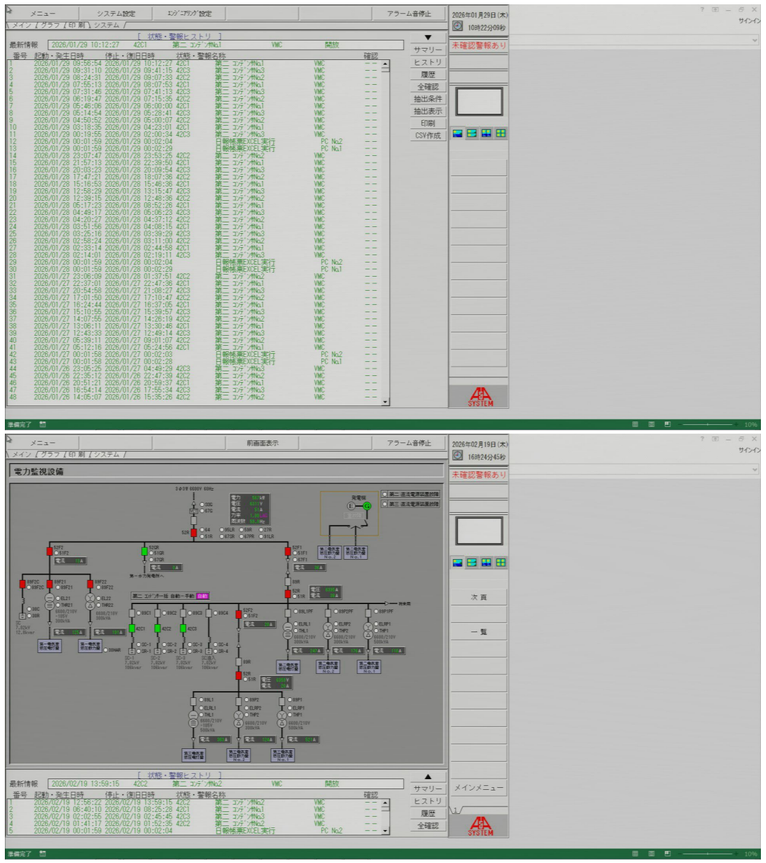
KVM Summary

- The company provides 4 KMV ports 9080, 9081, 9082, 9083
  - KVM 9080: show nothing
  - KVM 9081: **show different HMI SCADA– Hot Water System Overview (P&ID View) monitor scenes** which includes Piping & Instrumentation Diagram, used to monitor temperatures, flow rates, pumps, water levels, and power usage.
  - KVM 9082: This will **show 2 scenes** includes Alarm & Event Log Screen and Electrical Power Monitoring System
  - KVM 9083: Always show the Bath Temperature Control HMI
- Given provided APIs, we can only capture snapshots of KVM screens and move the mount. We **cannot** click/interact with the mount by API

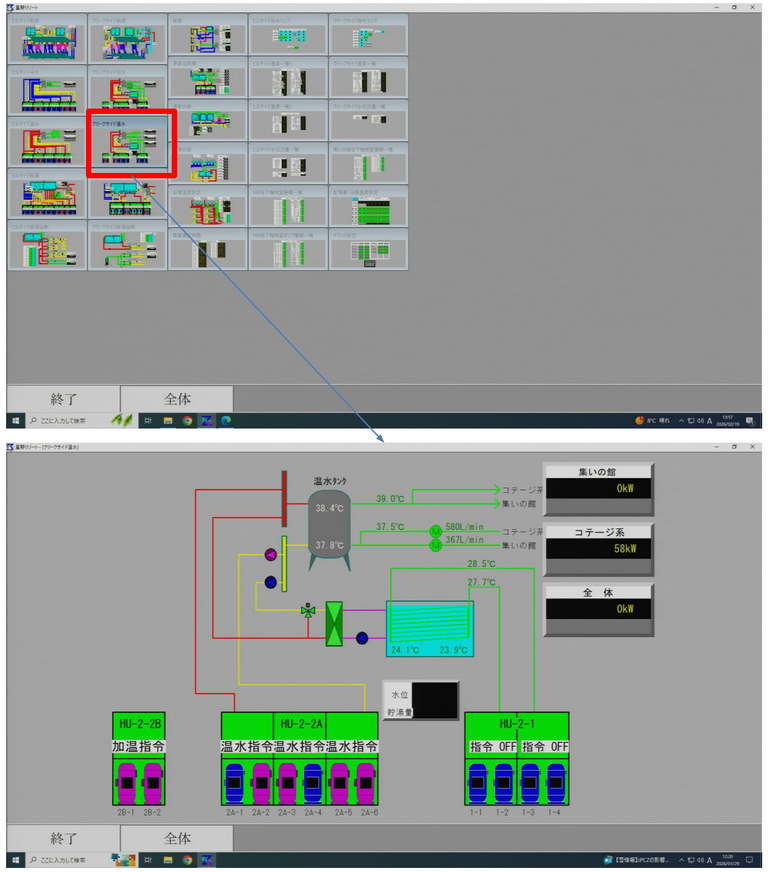


KVM 9080

KVM 9083



KVM 9082



KVM 9081

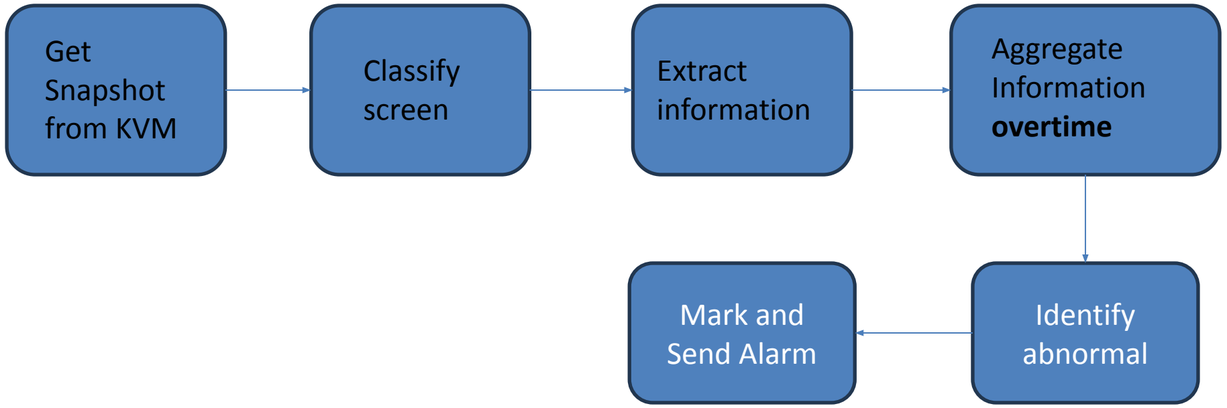
Company Requirement Summary

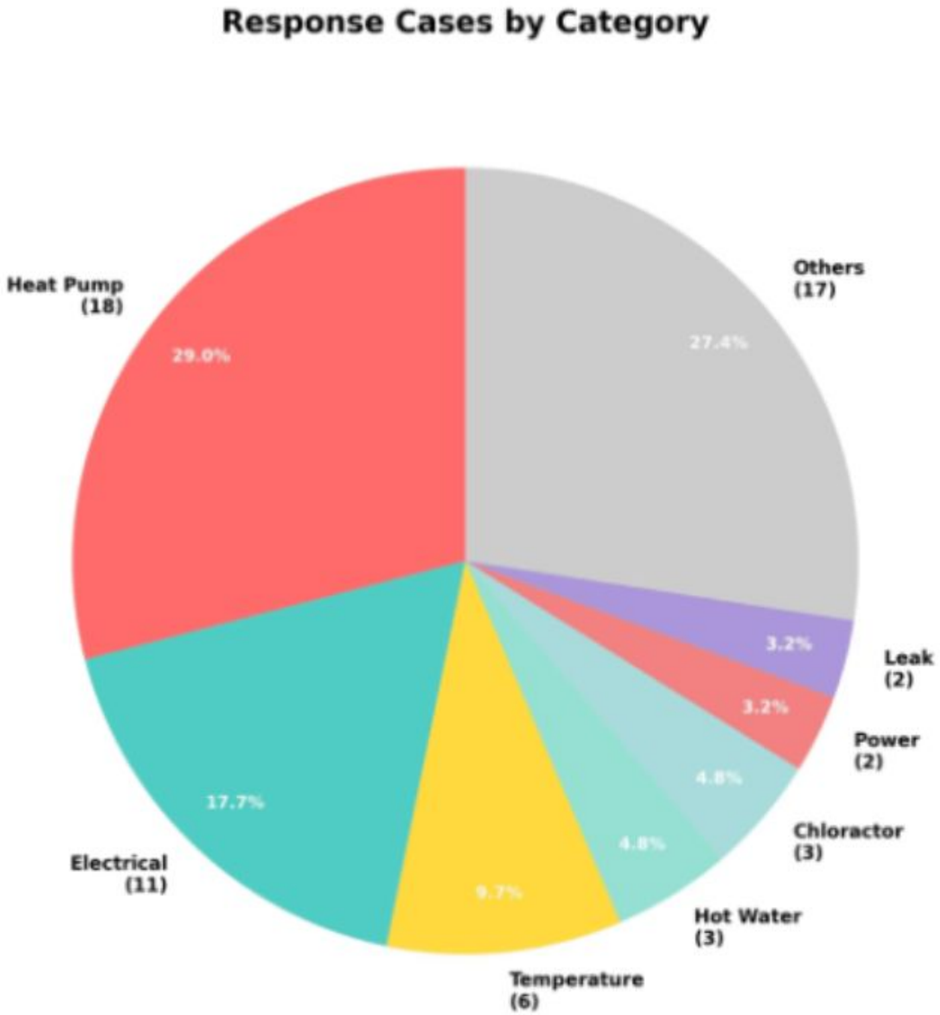
We need to find a way to detect whenever any abnormal event occurs by monitor KVM screens

The abnormal usually happens with heat pump and electrical services in night

We need to :

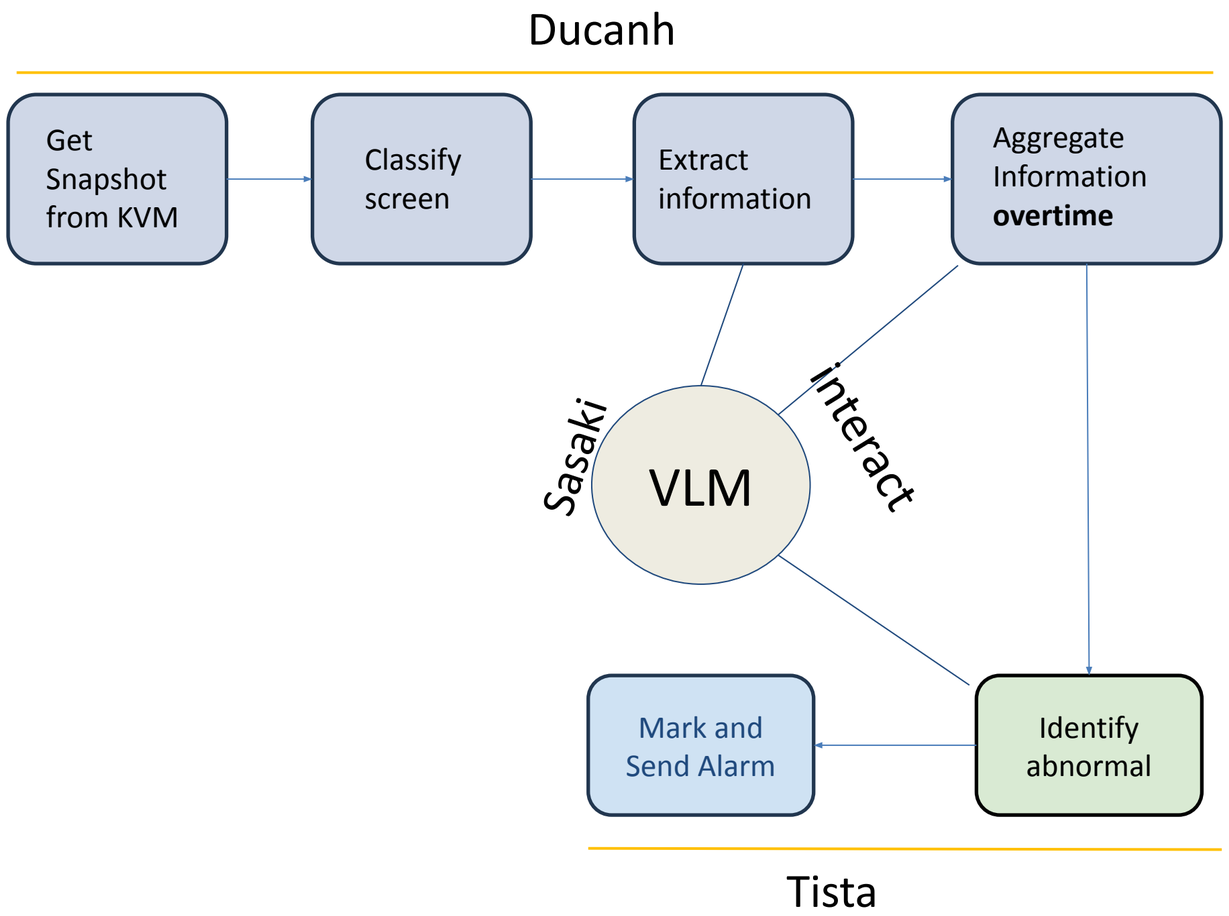
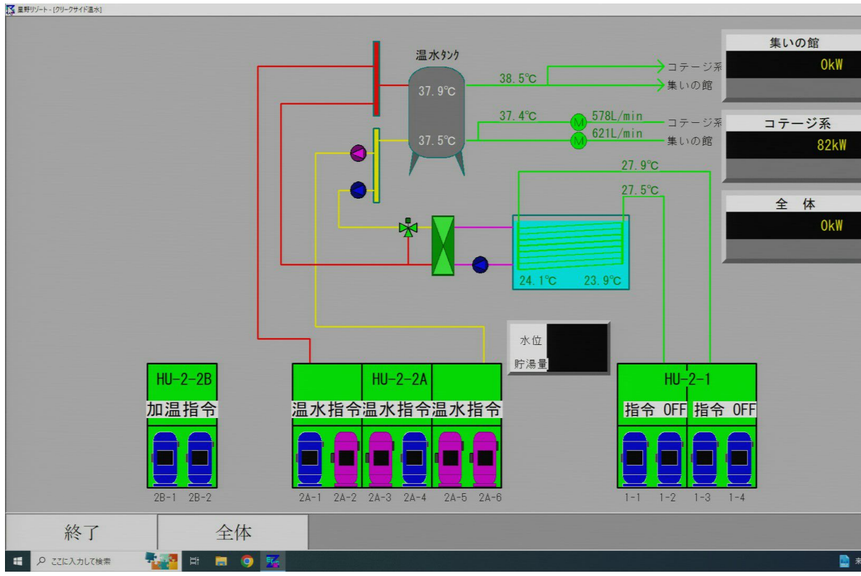
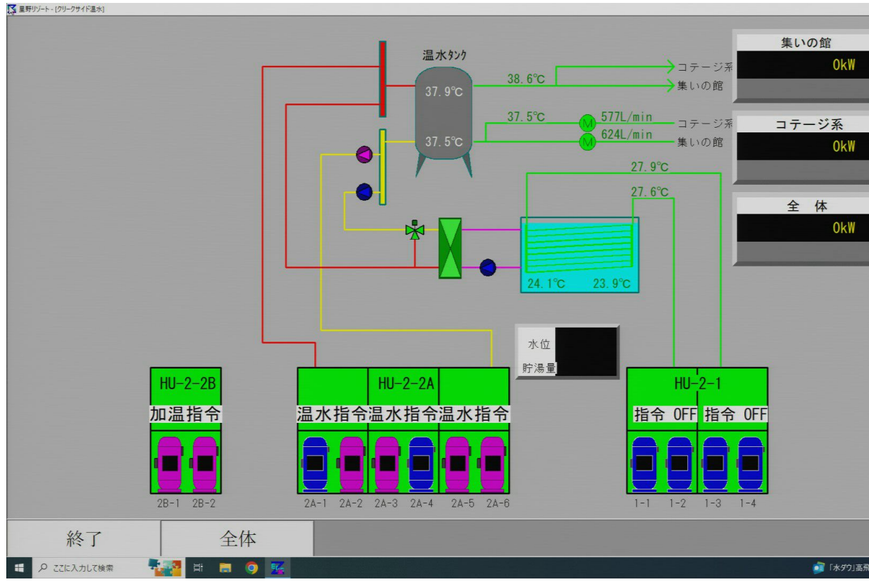
- **Identify the abnormal** and **which screen/system occur this event**
- For higher level, we could indicate the part/sensor in the system raise the error
- Send the alarm





Target:

Which information do we need to monitor?
In this spin, we will focus on monitoring temperature and water-flow values over time. Whenever any sensor changes, these values also change (as shown in the images).

Therefore, we need to implement and optimize the pipeline :

- **DucAnh** will implement basic pipeline for snapshot retrieval and information aggregation
- Sasaki will deploy and optimize VLM+prompt
- Tista will study solutions for abnormal event detection based on timeseries data